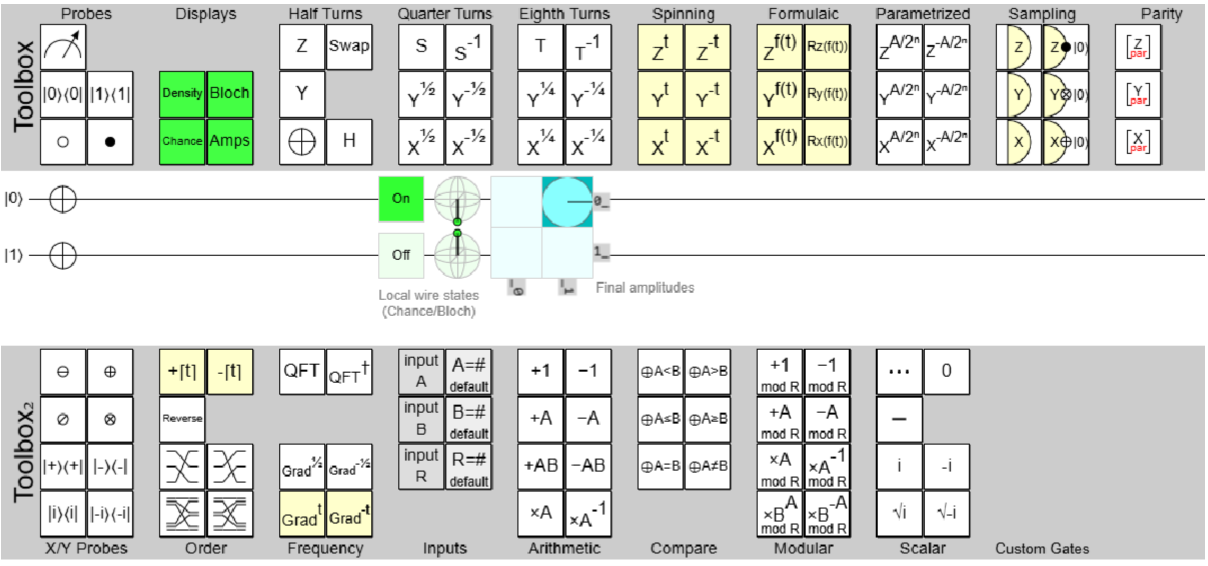
NAME: Manaswi Raj Branch / Sec: CSE / CSE -5

Batch: Lab 1

QUANTUM GATES USING SIMULATION

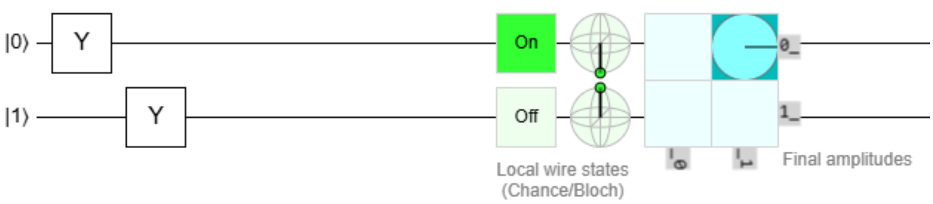
Pauli's X – Gate:



Truth Table:

INPUT	OUTPUT
|0>	|1>
|1>	|0>
α |0> +β |1>	α |1> +β |0>

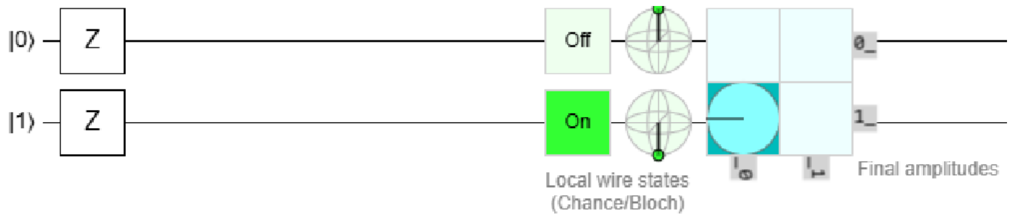
Pauli's Y – Gate:



Truth Table:

INPUT	OUTPUT
|0>	i|1>
|1>	-i|0>
α |0> +β |1>	α i |1> - i β |0>

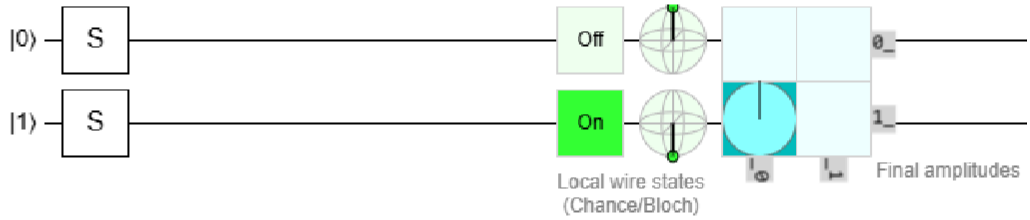
Pauli's Z – Gate:



Truth Table:

INPUT	OUTPUT
|0>	|0>
|1>	-|1>
α |0> +β |1>	α |0> - β |1>

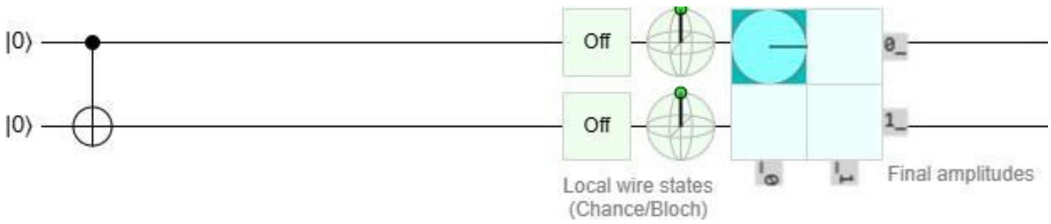
Pauli's S – Gate:



Truth Table:

INPUT	OUTPUT
|0>	|0>
|1>	i|1>
α |0> +β |1>	α |0> i β |1>

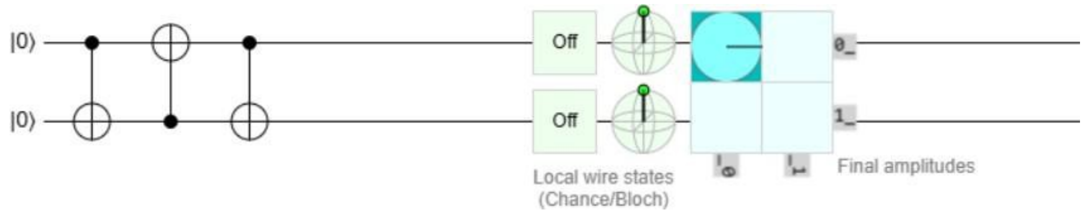
Controlled gate/CNOT gate



Truth Table:

INPUT	OUTPUT
|00>	|00>
|01>	|01>
|10>	|11>
|11>	|10>

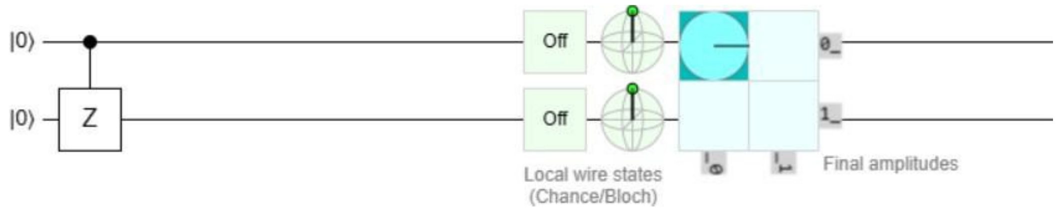
Swap Gate:



Truth Table:

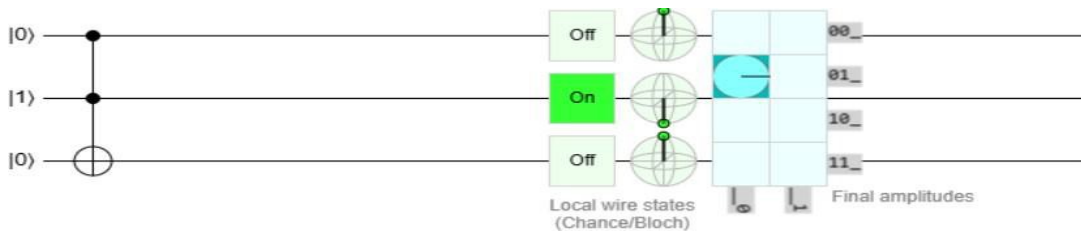
INPUT	OUTPUT
|00>	|00>
|01>	|10>
|10>	|01>
|11>	|11>

Controlled-Z gate:



INPUT	OUTPUT
|00>	|00>
|01>	|01>
|10>	|10>
|11>	-|11>

Toffoli gate (CCNOT, CCX, TOFF):



|a>	|b>	|c>	|a’>	|b’>	|c’>
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0